

metres are possible with 50kW laser systems. Again, preferred lasers are solid-state and fibre lasers.

Chemical oxy-iodine laser may also be used. For applications such as air defence against rocket, artillery, mortar targets, rocket-propelled grenades, battlefield missiles, laser-guided munitions and so on, typically, 100kW of laser power is needed for operational ranges of five to ten kilometres.

Long-range strategic applications of laser based DEW systems mainly include ballistic missile defence (Fig.



Fig. 4: Airborne laser for ballistic missile defence



Fig. 5: Space based laser concept in anti-satellite role

4), space control such as Space based lasers and anti-satellite applications (Fig. 5). In all these applications, operational ranges are generally in hundreds to thousands of kilometres, and required power levels are of the order of 1MW to 20MW depending upon the actual mission.

Space-control applications such as anti-satellite applications require relatively much higher power than that needed for ballistic missile defence.

Components of directed-energy laser weapon system

Unlike conventional military applications of laser systems such as laser range finders and target designators, which primarily comprise a laser source of desired specifications, laser based DEWs are much more than an HPL source as is evident from the block schematic of Fig. 6.

The sub-systems other than the HPL source are required for the purpose of directing the laser beam to the intended point on the target, keeping it there for the desired dwell time and producing the desired value of fluence on the target. Laser based DEWs essentially comprise two major sub-systems, namely HPL source and beam-control system.

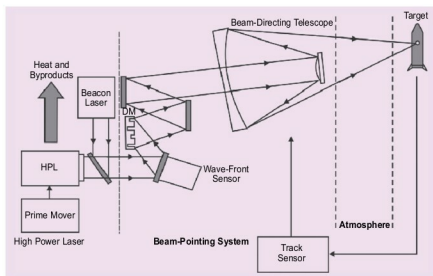


Fig. 6: Components of a laser based DEW system

An HPL source generates a laser beam of desired power level, beam quality and time duration to be able to inflict intended damage to the enemy target. Potential candidates for building HPL sources include gas dynamic CO₂ lasers, chemical lasers including hydrogen-fluoride (HF)/deuterium-fluoride (DF) and chemical oxy-iodine lasers, solid-state lasers and fibre lasers.

The ultimate quality desired of a potential HPL source is its ability to produce as high as possible a power density at the target for a given aperture size. Power density achievable for a given source-beam director combination is inversely proportional to square of wavelength and beam quality parameter for given beam director aperture size and operational range.

Therefore lower wavelength and smaller value of beam quality parameter signify its greater suitability as an HPL source. Smaller value of beam quality parameter indicates higher beam quality with unity being the ideal value. Shorter wavelength also has better target coupling efficiency.

In addition, power density at the target also depends upon atmospheric losses. Other important factors include power scalability, operator safety and ease of mounting.

Primary function of the beam-control system is to precisely point and focus laser energy at the designated point on the target and keep the same there for sufficient duration to cause intended damage to the target. In addition to target tracking and beam pointing and focusing, the beam-control system also has many other sub-systems to perform different tasks, which include beam shaping and beam stabilisation, wave-front control for atmospheric correction and so on.

In addition to laser source and beam-pointing system, propagation of an HPL beam through the atmosphere and its coupling with the target, also called target lethality, are the other important constituents determining